```
mysql> CREATE DATABASE College1DB;

Query OK, 1 row affected (0.03 sec)


mysql> USE College1DB;

Database changed

mysql> CREATE TABLE Department ( DeptID INT PRIMARY KEY, DeptName VARCHAR(50) NOT NULL, Location VARCHAR(50) );

Query OK, 0 rows affected (0.05 sec)


mysql> CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50) NOT NULL, Age INT, Gender VARCHAR(10), Marks DECIMAL(5,2), DeptID INT, Email VARCHAR(100), FOREIGN KEY (DeptID) REFERENCES Department(DeptID) );

Query OK, 0 rows affected (0.03 sec)


mysql> CREATE TABLE Course ( CourseID INT PRIMARY KEY, CourseName VARCHAR(50) NOT NULL, Credits INT );

Query OK, 0 rows affected (0.03 sec)


mysql> INSERT INTO Department VALUES (101, 'Computer Science', 'Block A'), (102, 'Pharmacy', 'Block B'), (103, 'Commerce', 'Block C'), (104, 'Mathematics', 'Block D');

Query OK, 4 rows affected (0.02 sec)

Records: 4  Duplicates: 0  Warnings: 0


mysql> INSERT INTO Student VALUES (1, 'Amit', 20, 'Male', 85.50, 101, 'amit@gmail.com'), (2, 'Priya', 21, 'Female', 92.00, 102, 'priya@gmail.com'), (3, 'Rahul', 19, 'Male', 76.50, 101, 'rahul@gmail.com'), (4, 'Neha', 22, 'Female', 88.00, 103, 'neha@gmail.com'), (5, 'Arjun', 20, 'Male', 69.50, 104, 'arjun@gmail.com'), (6, 'Sneha', 21, 'Female', 95.00, 102, 'sneha@gmail.com'), (7, 'Rohit', 23, 'Male', 72.00, 103, 'rohit@gmail.com'), (8, 'Pooja', 20, 'Female', 81.00, 104, 'pooja@gmail.com');

Query OK, 8 rows affected (0.01 sec)
```

Records: 8 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Course VALUES (201, 'Database Management System', 4), (202, 'Python Programming', 4), (203, 'Mathematics', 3), (204, 'Computer Networks', 3), (205, 'Data Structures', 4);

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> SHOW TABLES:

   -> \c

mysql> SHOW TABLES;

```
+----------------------+
| Tables_in_college1db |
+----------------------+
| course               |
| department           |
| student              |
+----------------------+
```

3 rows in set (0.02 sec)

mysql> select * from Department;

```
+--------+------------------+----------+
| DeptID | DeptName         | Location |
+--------+------------------+----------+
|    101 | Computer Science | Block A  |
|    102 | Pharmacy         | Block B  |
|    103 | Commerce         | Block C  |
|    104 | Mathematics      | Block D  |
+--------+------------------+----------+
```

4 rows in set (0.00 sec)

mysql> select * from Student;

```
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         1 | Amit        |   20 | Male   | 85.50 |    101 | amit@gmail.com  |
|         2 | Priya       |   21 | Female | 92.00 |    102 | priya@gmail.com |
|         3 | Rahul       |   19 | Male   | 76.50 |    101 | rahul@gmail.com |
|         4 | Neha        |   22 | Female | 88.00 |    103 | neha@gmail.com  |
|         5 | Arjun       |   20 | Male   | 69.50 |    104 | arjun@gmail.com |
|         6 | Sneha       |   21 | Female | 95.00 |    102 | sneha@gmail.com |
|         7 | Rohit       |   23 | Male   | 72.00 |    103 | rohit@gmail.com |
|         8 | Pooja       |   20 | Female | 81.00 |    104 | pooja@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
```

8 rows in set (0.00 sec)

mysql> select * from Course;

```
+----------+--------------------------+---------+
| CourseID | CourseName               | Credits |
+----------+--------------------------+---------+
|      201 | Database Management System |       4 |
|      202 | Python Programming       |       4 |
|      203 | Mathematics              |       3 |
|      204 | Computer Networks        |       3 |
|      205 | Data Structures          |       4 |
+----------+--------------------------+---------+
```

5 rows in set (0.00 sec)

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE ShowStudents()
   -> BEGIN
   -> SELECT * FROM Student;
   -> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> CALL ShowStudents();
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         1 | Amit        |   20 | Male   | 85.50 |    101 | amit@gmail.com  |
|         2 | Priya       |   21 | Female | 92.00 |    102 | priya@gmail.com |
|         3 | Rahul       |   19 | Male   | 76.50 |    101 | rahul@gmail.com |
|         4 | Neha        |   22 | Female | 88.00 |    103 | neha@gmail.com  |
|         5 | Arjun       |   20 | Male   | 69.50 |    104 | arjun@gmail.com |
|         6 | Sneha       |   21 | Female | 95.00 |    102 | sneha@gmail.com |
|         7 | Rohit       |   23 | Male   | 72.00 |    103 | rohit@gmail.com |
|         8 | Pooja       |   20 | Female | 81.00 |    104 | pooja@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
8 rows in set (0.01 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE ShowDepartments()
```

```
    -> BEGIN
    -> SELECT * FROM Department;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL ShowDepartments();
+--------+-----------------+----------+
| DeptID | DeptName        | Location |
+--------+-----------------+----------+
|    101 | Computer Science | Block A  |
|    102 | Pharmacy        | Block B  |
|    103 | Commerce        | Block C  |
|    104 | Mathematics     | Block D  |
+--------+-----------------+----------+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE ShowCourses()
    -> BEGIN
    -> SELECT * FROM Course;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL ShowCourses();
```

```
+----------+--------------------------+---------+
| CourseID | CourseName               | Credits |
+----------+--------------------------+---------+
|      201 | Database Management System |       4 |
|      202 | Python Programming       |       4 |
|      203 | Mathematics              |       3 |
|      204 | Computer Networks        |       3 |
|      205 | Data Structures          |       4 |
+----------+--------------------------+---------+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE GetStudent(IN sid INT)
    -> BEGIN
    -> SELECT * FROM Student WHERE StudentID = sid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL GetStudent(3);
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         3 | Rahul       |   19 | Male   | 76.50 |    101 | rahul@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE GetStudentsByDept(IN did INT)
    -> BEGIN
    -> SELECT * FROM Student WHERE DeptID = did;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL GetStudentsByDept(102);
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         2 | Priya       |   21 | Female | 92.00 |    102 | priya@gmail.com |
|         6 | Sneha       |   21 | Female | 95.00 |    102 | sneha@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
2 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE StudentsAboveMarks(IN m DECIMAL(5,2))
    -> BEGIN
    -> SELECT * FROM Student WHERE Marks > m;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> CALL StudentsAboveMarks(80);
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         1 | Amit        |   20 | Male   | 85.50 |    101 | amit@gmail.com  |
|         2 | Priya       |   21 | Female | 92.00 |    102 | priya@gmail.com |
|         4 | Neha        |   22 | Female | 88.00 |    103 | neha@gmail.com  |
|         6 | Sneha       |   21 | Female | 95.00 |    102 | sneha@gmail.com |
|         8 | Pooja       |   20 | Female | 81.00 |    104 | pooja@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE TotalStudents()
    -> BEGIN
    -> SELECT COUNT(*) AS TotalStudents FROM Student;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL TotalStudents();
+---------------+
| TotalStudents |
+---------------+
```

```
|           8 |
+---------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE CheckResult(IN sid INT)
    -> BEGIN
    -> SELECT StudentName, Marks, IF(Marks >= 50, 'Pass', 'Fail') AS Result FROM Student
WHERE StudentID = sid;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL CheckResult(3);
+-------------+-------+--------+
| StudentName | Marks | Result |
+-------------+-------+--------+
| Rahul       | 76.50 | Pass   |
+-------------+-------+--------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE GetStudentsByGender(IN gen VARCHAR(10))
    -> BEGIN
```

```
   -> SELECT * FROM Student WHERE Gender = gen;
   -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL GetStudentsByGender('Female');
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         2 | Priya       |   21 | Female | 92.00 |    102 | priya@gmail.com |
|         4 | Neha        |   22 | Female | 88.00 |    103 | neha@gmail.com  |
|         6 | Sneha       |   21 | Female | 95.00 |    102 | sneha@gmail.com |
|         8 | Pooja       |   20 | Female | 81.00 |    104 | pooja@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
4 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE StudentsBetweenMarks(IN m1 DECIMAL(5,2), IN m2
DECIMAL(5,2))
   -> BEGIN
   -> SELECT * FROM Student WHERE Marks BETWEEN m1 AND m2;
   -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> CALL StudentsBetweenMarks(70, 90);
```

```
+-----------+-------------+------+--------+-------+--------+-----------------+
| StudentID | StudentName | Age  | Gender | Marks | DeptID | Email           |
+-----------+-------------+------+--------+-------+--------+-----------------+
|         1 | Amit        |   20 | Male   | 85.50 |    101 | amit@gmail.com  |
|         3 | Rahul       |   19 | Male   | 76.50 |    101 | rahul@gmail.com |
|         4 | Neha        |   22 | Female | 88.00 |    103 | neha@gmail.com  |
|         7 | Rohit       |   23 | Male   | 72.00 |    103 | rohit@gmail.com |
|         8 | Pooja       |   20 | Female | 81.00 |    104 | pooja@gmail.com |
+-----------+-------------+------+--------+-------+--------+-----------------+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION AddTwoNumbers(a INT, b INT) RETURNS INT DETERMINISTIC
    -> BEGIN
    -> RETURN a + b;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT AddTwoNumbers(10, 20);
+----------------------+
| AddTwoNumbers(10, 20) |
+----------------------+
|                   30 |
+----------------------+
1 row in set (0.01 sec)
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION FindSquare(n INT) RETURNS INT DETERMINISTIC
   -> BEGIN
   -> RETURN n * n;
   -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT FindSquare(5);
+---------------+
| FindSquare(5) |
+---------------+
|            25 |
+---------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION ResultStatus(m DECIMAL(5,2)) RETURNS VARCHAR(10)
DETERMINISTIC
   -> BEGIN
   -> RETURN IF(m >= 50, 'Pass', 'Fail');
   -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT ResultStatus(75);
+------------------+
```

```
| ResultStatus(75) |
+------------------+
| Pass             |
+------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION StudentGrade(m DECIMAL(5,2)) RETURNS CHAR(1)
DETERMINISTIC
    -> BEGIN
    -> IF m >= 90 THEN RETURN 'A';
    -> ELSEIF m >= 75 THEN RETURN 'B';
    -> ELSEIF m >= 60 THEN RETURN 'C';
    -> ELSE RETURN 'D';
    -> END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT StudentGrade(85);
+------------------+
| StudentGrade(85) |
+------------------+
| B                |
+------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
```

```
mysql> CREATE FUNCTION AgeCategory(a INT) RETURNS VARCHAR(10) DETERMINISTIC
    -> BEGIN
    -> RETURN IF(a >= 18, 'Adult', 'Minor');
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT AgeCategory(20);
+-----------------+
| AgeCategory(20) |
+-----------------+
| Adult           |
+-----------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION CalculateBonus(m DECIMAL(5,2)) RETURNS INT DETERMINISTIC
    -> BEGIN
    -> IF m >= 90 THEN RETURN 1000;
    -> ELSEIF m >= 75 THEN RETURN 500;
    -> ELSE RETURN 200;
    -> END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT CalculateBonus(85);
+--------------------+
```

```
| CalculateBonus(85) |
+--------------------+
|                500 |
+--------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION HigherNumber(a INT, b INT) RETURNS INT DETERMINISTIC
    -> BEGIN
    -> RETURN IF(a > b, a, b);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT HigherNumber(10, 20);
+----------------------+
| HigherNumber(10, 20) |
+----------------------+
|                   20 |
+----------------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION DoubleMarks(m DECIMAL(5,2)) RETURNS DECIMAL(6,2)
DETERMINISTIC
    -> BEGIN
    -> RETURN m * 2;
    -> END //
```

```
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT DoubleMarks(85);
+-----------------+
| DoubleMarks(85) |
+-----------------+
|          170.00 |
+-----------------+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE FUNCTION AgeAfterFiveYears(a INT) RETURNS INT DETERMINISTIC
    -> BEGIN
    -> RETURN a + 5;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> SELECT AgeAfterFiveYears(20);
+-----------------------+
| AgeAfterFiveYears(20) |
+-----------------------+
|                    25 |
+-----------------------+
1 row in set (0.00 sec)

mysql> SELECT StudentName, Marks, StudentGrade(Marks) AS Grade FROM Student;
```

```
+-------------+-------+-------+
| StudentName | Marks | Grade |
+-------------+-------+-------+
| Amit        | 85.50 | B     |
| Priya       | 92.00 | A     |
| Rahul       | 76.50 | B     |
| Neha        | 88.00 | B     |
| Arjun       | 69.50 | C     |
| Sneha       | 95.00 | A     |
| Rohit       | 72.00 | C     |
| Pooja       | 81.00 | B     |
+-------------+-------+-------+
8 rows in set (0.00 sec)
```